เลขที่นั่งสอบ

College of Industrial Technology

King Mongkut's University of Technology North Bangkok

Midterm Examination of Semester 1 Year: 2018

Subject: 341151 Electric Circuits I Section: 5-6

Date: 4 October 2018 Time: 15.00-17.00

Name:______________________________ID:________________Class:__________________

Instructions:

1. The examination has 12 pages (including this page), 3 sections (49 questions) and a total score of 65 points.
2. Write all your solutions and answers on this examination sheet.
3. This is a closed book examination.
4. You are not allowed to leave the exam room during the first 1 hour after the beginning of the exam.
5. You are not allowed to open the exam papers or start to answer before the proctor's permission.
6. You are not allowed to use the restroom during the exam except in case of an emergency.
7. No documents are allowed to be taken out of the examination room.
8. Calculators is allowed in the examination.
9. Electronic communication devices are NOT allowed in the examination room.
10. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

Part 1 (15 points) True/False

Instruction: Mark the correct answer for following questions in the provided answer sheet.

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1. A potential difference is always measured at one point.
2. The prefix that corresponds to a power of ten of 3 is Mega.
3. Conductance (G) is the same as resistance (R).
4. The efficiency of a system is the ratio of the power output divided by the power input.
5. Kirchhoff's Voltage law states that for a closed loop series path, the algebraic sum of all the voltages around any closed loop in a circuit is equal to zero.
6. The current is the maximum through all elements of a series circuit.
7. For good conductors, a decrease in temperature will result in an increase in the resistance level.
8. Kirchhoff's Current law states that the total current entering a circuits junction is exactly equal to the total current leaving at the same junction.
9. Kirchhoff's voltage law requires that total voltage input to a system must equal the total voltage output from the system.
10. Alessandro Volta invented the first battery.
11. One square mil equals to $4/\pi$ CM.
12. AWG stands for the American Wire Gauge, which is used to indicate the cable weight.
13. The electrical resistivity of a copper cable is the smallest.
14. The heat from a microwave oven occurs due to the friction of the water molecules in the food.
15. Grounding is not important. We do not need to have the grounding in the house.

Part 2 (30 points) Multiple choices

Instruction: Mark the correct answer for following questions in the provided answer sheet.

1. Four 12-volt batteries connected together in parallel will result in which of the following voltage?

   (a) 3 volts (b) 6 volts

   (c) 12 volts (d) 48 volts

2. Current is defined as the __________pass though the imaginary plane.

   (a) charge (b) voltage

   (c) ampere (d) current

3. The current divider rule states that for two series ______________ of equal value, the current will divide equally.

   (a) elements (b) resistors

   (c) circuits (d) branches

4. A voltmeter is always placed _____________ an element to measure the voltage.

   (a) in parallel with (b) after

   (c) in series with (d) on top of

5. Express the number 0.000003597 A using the SI perfix.

   (a) 359 mA (b) 3.59 μA

   (c) 35.9 pA (d) 359.7 nA

6. Find voltage between two points if the energy of 50J is required to move a charge of 10 C between two points.

   (a) 500 V (b) 50 V

   (c) 5 V (d) 0.5 V

7. Which of the following expresses a double-superscript notation of V_{ab} = 15 V?

(a) Point a has a higher potential than b.

(b) Point a has the same potential as b.

(c) Point a has a lower potential than b.

(d) Point a cannot be determined

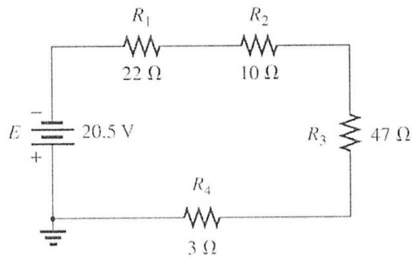


Figure 1.

8. See Figure 1. The total resistance in this circuit is

(a) 82 Ω (b) 22 Ω

(c) 100 Ω (d) 10 Ω

9. See Figure 1. The total current flowing from the battery is

(a) 2.5 mA (b) 0.25 mA

(c) 2.5 A (d) 0.25 A

10. See Figure 1. The total power dissipated by the 47 Ω resistor is

(a) 11.75 W (b) 2.93 W

(c) 1.33 mW (d) 8.94 W

11. The sum of the series voltage drops around a closed loop must ________.

(a) be equal to infinity (b) be equal to the applied voltage

(c) be equal to zero (d) be the same

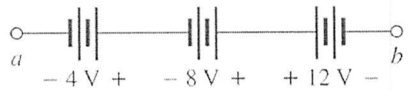


Figure 2.

12. See Figure 2, find a single voltage source between points a and b.

(a) 24 V (b) 16 V

(c) 8 V (d) 0 V

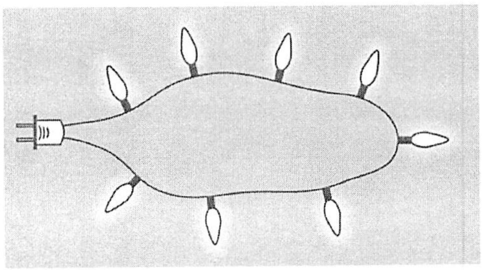


Figure 3.

13.See Figure 3. Eight holiday lights are connected in series as shown in Figure 3. If the set is connected to a 220 V source and each bulb has an internal resistance of 28 Ω, determine the power delivered to each bulb.

(a) 24 W (b) 27 W

(c) 27.5 W (d) 20 W

14.See Figure 3. What is the voltage drop across the bulb?

(a) 24 V (b) 27 V

(c) 27.5 V (d) 20 V

15.If one bulb burns out, what is the effect on the remaining bulb?

(a) Nothing happened (b) All go out

(c) All burn out (d) All turn on

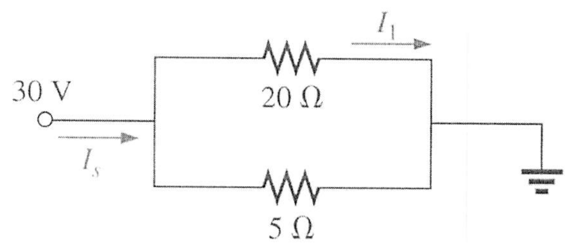


Figure 4.

16.See Figure 4. What is the current I_S?

(a) -7.5 A (b) +7.5 A

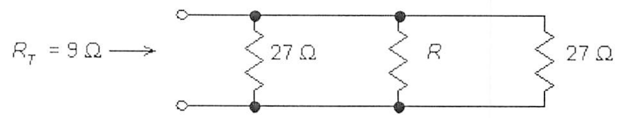
(c) -1.2 A (d) +1.2 A

17.See Figure 4. What is the current I_1?

(a) -6.0 A (b) +6.0 A

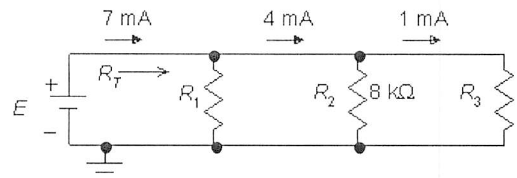
(c) -1.5 A (d) +1.5 A

18.Find the value of R.



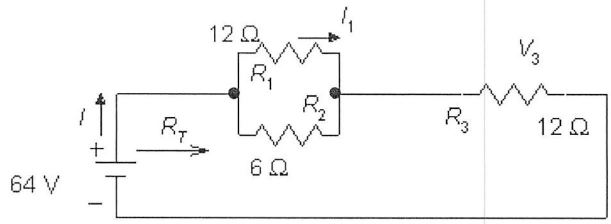
(a) 9 Ω (b) 54 Ω (c) 18 Ω (d) 27 Ω

19.Find R_1.



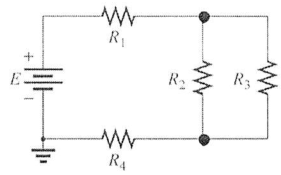
(a) 11 kΩ (b) 12 k Ω (c) 9 kΩ (d) 8 kΩ

20.Calculate R_T.



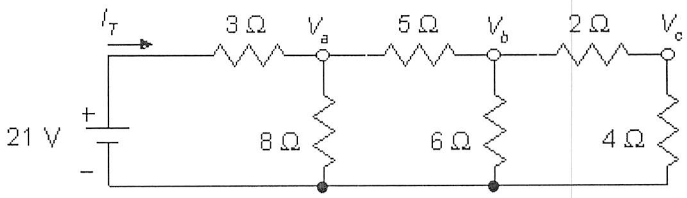
(a) 30 Ω (b) 16 Ω (c) 24 Ω (d) 18 Ω

21.Find the value of R_1. Given R_2=2Ω, R_3=2Ω, R_4=5Ω and R_T=10Ω.



(a) 9 Ω (b) 4 Ω (c) 8 Ω (d) 20 Ω

22.Find I_T.



(a) 3 A (b) 3.5 A (c) 2.5 A (d) 4 A

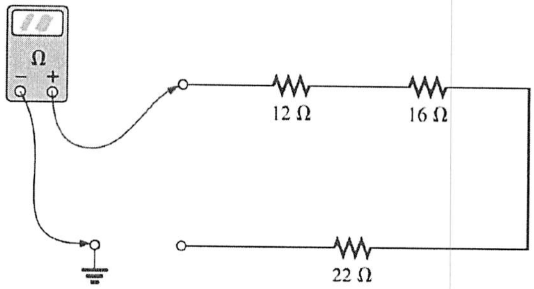
23.How many dissipated power (W) is there at a 5 Ω resistor connected across a 10V battery?

(a) 20W (b) 2W (c) 10W (d) 1W

24.What is the power deliver to a resistor 1 kΩ with current 3 mA?

(a) 9 mW (b) 9 nW (c) 90 mW (d) 3 W

25.From figure below, what is the ohmmeter reading?



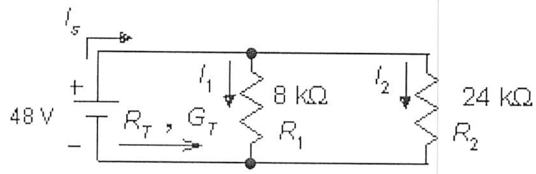
(a) 0 Ohms

(b) 50 Ohms

(c) 12 Ohms

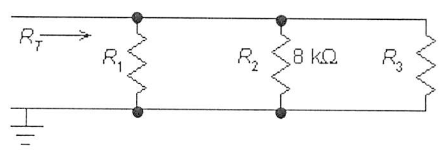
(d) Infinity Ohms

26.Find the total conductance.



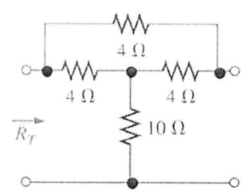
(a) 166.7 μS (b) 1.667 S (c) 1.667 mS (d) 16.7 S

27.From figure below, what is R_T? Given R_1=R_3=2Ω



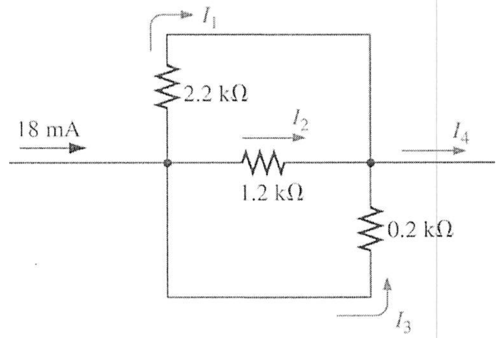
(a) 0.88Ω (b) 0.99 kΩ (c) 1 kΩ (d) 1.88 kΩ

28.Calculate R_T.



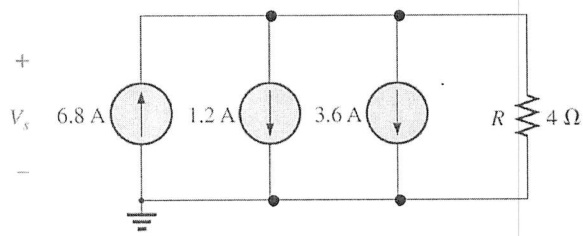
(a) 14 Ω (b) 12.6 Ω (c) 24 Ω (d) 18 Ω

29. Following the circuit below, find the current I_4.



(a) 18.3 mA (b) 1.8 mA (c) 18 mA (d) 18.5 mA

30. Determine the source voltage V_s

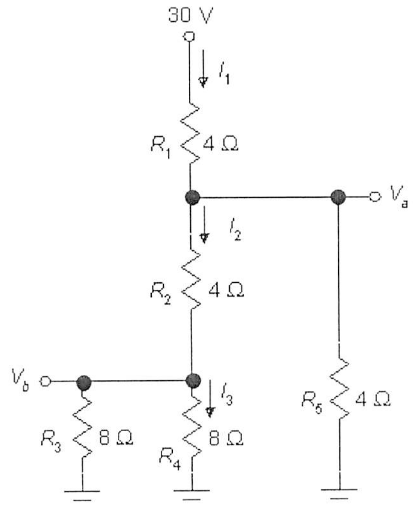


(a) 8 V (b) -8 V (c) 46.4 V (d) -46.4 V

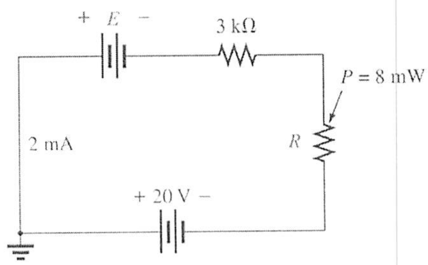
Part 3 (20 points) Calculation

Instruction: Show the mathematical expression and answers of following problems.

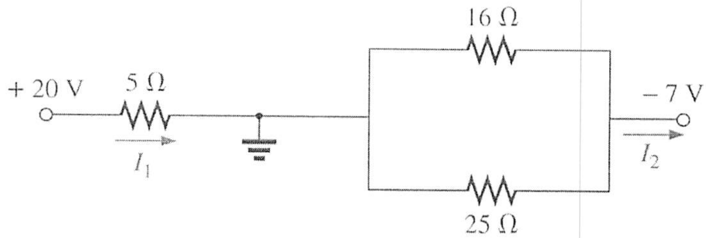
1. (5 points) From Figure below, determine the current I_1, I_2, I_3, and voltage V_a , V_b



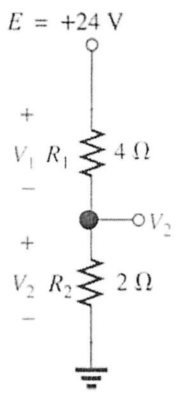
2. (5 points) Find the unknown voltage source and resistor R for the networks below. Also indicate the direction of the resulting current.



3. (5 points) Determine I_1 and I_2 from the following circuit.



4. (5 points) Determine the value of V_2 and V_2 by using voltage divider.



Answer Sheet

Name:______________________________ ID:______________

Subject: 341151 Electric Circuits I

Part 1

	True	False
1.		
2.		
3.		
4.		
5.		
6.		
7.		
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Part 2

	a	b	c	d
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